



Department of Computer Science and Engineering
Final Examination Summer 2025
CSE 321: Operating Systems

Duration: 2 Hours

Total Marks: 50

Answer all the following questions.
 Figures in the right margin indicate marks.

Name:	ID:	Sec:
--------------	------------	-------------

1. a) Write the answer in the question paper.

[0.5*

CO1 Find outputs of the given code:

6=3]

```

2  int direction = 0;
3  int array[] = {11,22,33};
4  int high;
5  int low;
6  void searchForIndex(int value);
7  int main(void) {
8      int count = sizeof(array) / sizeof(int);
9      high = count - 1;
10     low = 0;
11     searchForIndex(33);
12     return 0;
13 }
14 void searchForIndex(int value) {
15     while (1) {
16         printf("Current search range: low = %d, high = %d\n", low, high);
17         if (high < low) {
18             exit(0);
19         }
20         int mid = (high + low) / 2;
21         if (array[mid] == value) {
22             printf("Value %d found at index %d\n", value, mid);
23             exit(mid);
24         }
25         else {
26             pid_t pid = fork();
27             if (direction % 2 == 0) {
28                 if (pid < 0) {
29                     perror("fork failed");
30                     exit(1);
31                 }
32                 else if (pid > 0) {
33                     wait(NULL);
34                     high = mid - 1;
35                 }
36                 else {
37                     low = mid + 1;
38                 }
39             }
40             else {
41                 if (pid < 0) {
42                     perror("fork failed");
43                     exit(1);
44                 }
45                 else if (pid > 0) {
46                     wait(NULL);
47                     low = mid + 1;
48                 }
49                 else {
50                     high = mid - 1;
51                 }
52             }
53             direction = direction + 1;
54         }
55     }
56 }
57

```

Write the outputs in the question paper. The output sequence should be exactly matched.

Outputs	
1	
2	
3	
4	
5	
6	

b) Write the answer in the answer script.

[1*2
=2]

A running process calls a function **compute()**, which declares several local variables. After **compute()** returns, the process calls another function, **display()**.

- I. **Identify** in which section of the PCB are the local variables for **compute()** stored?
- II. **State** what happens to this field when the **compute()** function is called and when it returns?

2. Write the answers in the answer script. [1*2
CO2 =2]
- a) Alice has designed an OS which is mounting vsfs file system. After reading the superblock, it knows the total number of data blocks and the starting block of the inode table.
- Write** what are the next two fields the OS needs to read to manage the allocation of space?
 - State** the functions of these fields?
- b) Bob changes the permissions of a file using the **chmod** command. He notices that the file's "**last modified**" timestamp remains unchanged. [1*2
=2]
- Identify** which specific field of the file or the file system is being altered?
 - In this scenario, **why** doesn't this operation count as modifying the file's content?
- c) A file system uses UNIX inode data structure which contains **24** direct block addresses, **40** single indirect blocks, some double indirect blocks, and **5** triple indirect blocks. The size of each block is **128 Bytes** and the size of each pointer is **32 Bytes**. The maximum possible file size is **87 KB**. [1*2
=2]
- Calculate** how many pointers can be stored in a block in this file system?
 - Calculate** the number of double indirect blocks in the system.
- d) Dipu removed a file from the disk which causes the deletion of the last remaining hard link to a file. [1*2
=2]
- Detect** which two types of resources are now marked as free by the file system?
 - State** which fields of the file system are updated to reflect this change?
- e) A file system has inode size = **1 KB** and block size = **8 KB**. First **5 blocks** contain superblock, journal, group descriptor, data bitmap and inode bitmap. [1+2
=3]
- Calculate** the address of the inode number **19**.
 - The disk is sector addressable and the size of each sector is **512 Bytes**. **Find out** the sector address of the inode block.
- f) A program creates a new file **/tmp/new.txt**. [0.5*
4=2]
- Write down four mandatory** write operations that are required just to create the empty file in its parent directory **/tmp**.

g) Shoumo wants to optimize journal writes by issuing all five blocks of a transaction (**TxB**, **I[v2]**, **B[v2]**, **Db**, **TxE**) to the disk in a single large request for speed. **[1*2 =2]**

- I. **Identify** why is this approach unsafe if the disk reorders writes internally?
- II. **What** incorrect data might be replayed during recovery if a crash occurs at the wrong time according to the scenario?

h) Dipto is a system administrator. To improve performance, he switches the mechanism of the journaling to metadata journaling in his system. **[1*2 =2]**

- I. **Identify** the key difference between data journaling and metadata journaling?
- II. **State** the critical write ordering principle maintained by metadata journaling to ensure the consistency.

i) Kabbya executes an application which deletes a directory, freeing its data block (block #123). Immediately after, it creates a new file which reuses that same block #123 for its own data. A crash occurs before the transaction involving the directory deletion is cleared from the journal. **[1*2 =2]**

- I. **State** what issue can arise during the recovery procedure when it replays the old transaction for the directory according to the scenario?
- II. **What** type of record does **ext3** add to the journal to prevent this problem?

j) A web browser application needs to download images from a network while simultaneously allowing the user to scroll through the text that has already loaded. **[1*2 =2]**

- I. **Identify** which feature of modern operating systems allows a single process to perform these two tasks concurrently?
- II. **State** how would multithreading be applied in this scenario?

k) An algorithm is designed to process a large array. It works by having **four threads** run on **four separate cores**, with each thread processing **one-quarter** of the array elements using the exact same logic. **[1*2 =2]**

- I. **Identify** what type of parallelism is functional in the application?
- II. **State** about the defining characteristic of this type of parallelism.

l) A developer is using semaphores with waiting queues to manage resources. He is concerned that his design might lead to a situation where two or more processes are waiting for an event that can only be caused by one of the other waiting processes. [1*2
=2]

- I. **Identify** the potential issue in this scenario?
- II. In the context of semaphores, **state** that what is the specific "event" that the processes are waiting for?

3. **Write the answers in the answer script.**

CO3

a) A C++ program uses a pointer that holds the address **0x400**. When the program runs, the hardware actually accesses memory location **0x18400**. [1*2
=2]

- I. **Identify** specifically which two types of addresses are involved in this scenario?
- II. **State** what hardware unit is responsible for this conversion.

b) A process with a memory requirement of **145698 Bytes** is running on a system. **18 pages** are required to allocate the process in the system. [1*2
=2]

- I. **Calculate** the page size of the system .
- II. **Calculate** the **internal fragmentation** of the process after paging.

c) A process attempts to access a memory address that belongs to page number **7**. The system's page table has the information of frame numbers of the pages from **0** to **5**, but not for **6** and **7**. [1*2
=2]

- I. **State** which attribute within the page table is used to identify the illegal page access?
- II. **Write** how the OS uses this mechanism to differentiate between legal and illegal pages for a process?

d) In a **32-bit** system with a **two-level** page table, the logical address is split into a **9-bit outer page number**, a **9-bit inner page number**, and a **14-bit offset**. [1*2
=2]

- I. **Calculate** how many entries can the inner page table hold in this system?
- II. **State** what does each entry in the inner page table point to?

e) A process starts, and the OS sets the instruction pointer to the first instruction. This instruction is on a page that is not yet in the physical memory. [1*2
=2]

- I. **State** what is this technique of loading pages only when they are needed to be executed by the CPU?
- II. **Identify** what is the immediate consequence of the process trying to execute this first instruction according to the scenario given above?

f) A system using the **FIFO** page replacement algorithm experiences a page fault but has no free frames. The pages currently in memory were loaded in the order: P1, P2, P3. [1*2
=2]

- I. **Identify** which page will be replaced in this scenario.
- II. **State** which data structure is used to implement this algorithm?

g) Consider a system with TLB miss ratio of **4.25%**, and the main memory access time of **125 ns**. If Effective Access Time is **142.3125 ns**, then **calculate** the TLB access time. [1*2
=2]

h) A system is processing the reference string **7, 0, 1, 2, 0, 3, 0, 4,** with three frames, which currently hold pages **{7, 0, 1}**. The next reference is to **page 2**, requiring a replacement. The upcoming references are **....., 0, 3, 0, 4,** [1*2
=2]

- I. If the system uses **Optimal** page replacement algorithm and the CPU demands for **page 2** then find out which page should be swapped out in order to swap in the demanded page in this scenario?
- II. **Justify** your selection based on the future reference strings.

- i) A process runs in a system with **3-level** paging and it has a logical address space of **16-bits**. In the system page size is **64 Bytes**, size of each entry of the page table is **4 Bytes** and size of the main memory is **2 KB**. After the paging the logical address space of the process looks like:

[2*2
=4]

2 nd Outer Page #	Outer Page #	Inner Page #	Offset
P1	P2	P3	d
2	4	4	6

Necessary page table information is given below:

2 nd Outer PMT		
Page #	Frame #	v/i bit
0		0
1	27	1
2	13	1
3		0

Outer PMT						Inner PMT					
Frame 13			Frame 27			Frame 2			Frame 9		
Page #	Frame #	v/i bit	Page #	Frame #	v/i bit	Page #	Frame #	v/i bit	Page #	Frame #	v/i bit
0			0			0			0		
1			1			1	6	1	1	39	1
2			2			2			2		
3	99	1	3			3			3		
4			4			4			4		
5			5	2	1	5			5		
6			6			6			6		
7			7			7			7		
8	9	1	8			8			8		
9			9	22	1	9	125	1	9		
10			10			10			10		
11			11			11			11		
12			12			12			12	59	1
13			13			13			13		
14			14			14			14		
15			15			15			15		

In this system, if the CPU generates logical addresses I) **22135** and II) **41046** then map the corresponding physical addresses of these logical addresses. If the physical address is not valid then write “invalid”.